

Seminar Report on Express: Lowering the Cost of Metadata-hiding Communication with Cryptographic Privacy

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Abstract

This report studies the reproducibility and performance of Express, a system for metadata-hiding communication. Using two servers, express ensures cryptographic security against at most one malicious server and an arbitrary number of malicious clients. The paper presents new protocols that offer low communication overhead, regardless of the number of users, and improved computation costs using symmetric key cryptography. Compared to Pung and Riposte, two previous cryptographically secure systems, Express achieves increased message throughput, while maintaining strong privacy guarantees. This makes Express an ideal applications for communication with whistleblowers.

1 Introduction

In today's digital landscape, secure communication has become increasingly crucial, with growing capabilities of surveillance and metadata analysis. Secure messaging apps like Signal and Transport Layer Security (TLS) encrypt the content of sent messages, but they do not protect metadata. It is possible to observe who sent and received messages, when and how often they communicated and how much data they exchanged. These information are as sensitive as the message itself and should therefore not be leaked. It is challenging to keep all metadata hidden, as most services for communication over the internet rely on some of them. An option for metadata hiding would be using a TOR browser or SecureDrop, but they are vulnerable to attacks by adversaries that control many network points. Other existing systems to hide metadata either have to high communication cost to be used in reality, or they have low computation cost but their security gets worse with every round of communication, also making them impractical.

While two servers are required, express still runs with one malicious server (Eskandarian et al., 2021). Express' target audience are whistleblowers. To receive messages, a journalist or any other person that should receive information registers a mailbox with the two servers. The server assigns a 128-bit virtual address to each mailbox and maintains a page table in which it stores the corresponding physical address. The client can transmit this address to the whistleblower. Now the client can upload a message into the mailbox. No one except the mailbox owner will know that a message has been sent into this mailbox. Mailbox owners have to check their mailboxes regularly. Express does not hide when a client read from a mailbox, therefore a pattern in the interval of checking could reveal something about activity of this mailbox. Additionally, Express must implement a possibility for cover traffic. This provides whistleblowers with plausible deniability and ensures that they cannot be accused of giving information simply for contacting an Express Server.

There are different types of attacks that Express must be resilient against. All clients can read from all mailboxes, so Express must re-encrypt the mailboxes differently each time a read happens, providing a encryption only the owner can decrypt. All others can neither read the content nor learn anything about which client is writing, even when reading the content of all mailboxes between every pair of writes. Another possible threat would be a malicious client targeting all mailboxes. It could overwrite all mailboxes with one message. Because the mailboxes can only hold a maximum of one message, all messages sent before this will be lost due to this denial-of-server attack. The complete overview is divided between the two servers, which means that the servers alone cannot recognise that this is not a valid message. A new auditing protocol presented in the paper is used to prevent such attacks. However, a malicious client could also attack just one mailbox, in which case the message would be completely correct and the server would have no way of detecting and preventing the attack. Virtual addresses are used to prevent the original messages in the mailbox from being overwritten. The address range is so large that the probability of hitting a specific mailbox is negligible.

2 Background

This section explain two key concepts that form the foundation of Express: cryptographic privacy and distributed point functions, also called DPFS. In the following section, we will show how they can be used to achieve efficient metadata-hiding communication.

2.1 Cryptographic privacy

Cryptographic privacy is a mechanism that guarantees no information about communication metadata leaks to an adversary. Even after observing many rounds of communication, the adversary cannot determine which clients are communicating with each other or which mailbox a message was written to.

An alternative mechanism is differential privacy. After several rounds of communication, systems based on differential privacy such as Vuvuzela or Stadium, will lose security. Consistent observation will lead to some leaked information. Therefore, although cryptographic privacy is slower, it is the preferred approach for a frequently used applications.

2.2 Distributed point function

A DPF (<https://eprint.iacr.org/2023/028.pdf>) is a cryptographic algorithm. It splits a function f into function shares that individually hide f . The point function $f_{a,b}(x)$ outputs b at the point $x = a$ and 0 everywhere else. The two or more parts of the function individually reveal nothing about the function.

Express splits the function into two parts, one for each of the servers. Because no server alone can reconstruct the full function, even a malicious or compromised server cannot tell anything about the function and learns nothing about which mailbox a client wrote into.

3 Private Communication in Express

In this section, we will explain how an Express client writes messages into mailboxes and how the servers store them.

3.1 Express Servers

Secret sharing with DPFs. If the express servers hold n mailboxes together and each of them contains an element of the finite field $\mathbb{F}$, the totality of all mailboxes can be represented as a vector: $D \in \mathbb{F}^n$. D can therefore be represented as the sum of the two components $D_A + D_B = D$, where the vector belonging to server A corresponds to $D_A \in \mathbb{F}^n$ and server B's vector corresponds to $D_B \in \mathbb{F}^n$. When the servers receive a new message w , they can both add this vector to their current status: $D_A \leftarrow D_A + w_A \in \mathbb{F}^n$ and equally $D_B \leftarrow D_B + w_B \in \mathbb{F}^n$.

By using DPFs, as explained in the following subchapter, the servers do not receive a complete vector, but only a function share. They can evaluate this at any point, i.e. for each mailbox. This gives them the vectors, which they can add to D_A and D_B as before.

Distribution of trust. To prevent one server from containing all the data, thus representing a single point of failure and not providing adequate security for metadata hiding, multiple servers must be used. One way to achieve this distribution of trust would be to use a large number of servers. This offers the advantage that even if more than one server is malicious, the system can still continue to run.

Express uses only two servers, which in the paper are the Wall Street Journal and the Washington Post. These two must therefore meet stricter security requirements. One difficulty in implementing Express could be censorship. If access to one of the two servers is blocked, the entire application will no longer function. Express is less suitable for countries where news websites, i.e. servers on which Express runs, are often restricted.

Malicious server Another danger is posed by a malicious server. A denial of service attack can originate not only from a client, but also from a server. The server can overwrite the contents of all mailboxes with meaningless content, causing the original content to be lost. However, since the client will realise that the contents of their mailbox are garbage, they can recognise that the system can no longer be trusted.

3.2 Express Clients

If a client wants to write a message $m \in \mathbb{F}$ to a mailbox, it can insert it into a vector of the length of all mailboxes by multiplying it with the i -th standard-basis vector $\mathbf{e}_i$. This vector should be zero everywhere except at the location of the targeted mailbox. Mathematically, this can be formulated as follows: $m \cdot \mathbf{e}_i \in \mathbb{F}^n$. This vector must then be split so that one can be sent to each server; i.e. $w_A + w_B = m \cdot \mathbf{e}_i$. However, the costs of these calculations are high. For every message that the client sends, it must have an element in the vector for each mailbox in the system and send it to both servers. A better alternative is to use DPFs, which allows clients to send less data. As explained in section 2.2, it is possible to split a function into two parts that, on their own, do not reveal anything about the function. We call the functions for the two Servers A and B f_A and f_B . When added together, they give exactly the value of f at each point. This point function can therefore also be written as follows: $f_{i^*,m} : [N] \rightarrow \mathbb{F}$. i^* is the only point at which this function does not yield zero, but rather m . Clients can now generate the shares f_A and f_B . Added at any point $i \in [N]$ they result in the DPF: $f_A(i) + f_B(i) = f_{i^*,m}$. Individually, these do not tell an attacker anything about i^* , i.e. the mailbox being written to, or about m , i.e. the message being transmitted.

The use of DPFs has reduced the size of sent messages to such an extent that, with a security parameter λ , each part of the function now has a length of $O(\lambda \log N + |m|)$ bits.

The messages sent by clients should either all be the same size or consist of a predefined set of sizes. If an attacker can extract information from the message size, the system loses security and the cover traffic no longer provides sufficient plausible deniability.

4 Protecting Privacy and Integrity

In this section we first show how Express ensures privacy when all clients behave legitimately, and then when at least one is malicious. At the end, we will show how Express can be used as a tool for communication between whistleblowers and journalists.

4.1 Privacy with honest clients

The previous chapter (3) explained how we can write something secretly to the server. However, an adversary could now check all network traffic. If they compare the contents of all mailboxes after each write, they will not be able to read the message because it is encrypted, but they can find out which client wrote to which mailbox.

To ensure that the metadata is completely hidden, we also need to control who reads the mailboxes. This could be prevented by only allowing reading once all clients have written, i.e. by running the system in batches. Pung and Riposte are based on this concept. However, this severely limits actual usability. Express has therefore proposed a method that does not require synchronised rounds. To achieve this, we must ensure that an attacker cannot view the messages or identify which mailbox has been modified after a message has been sent. The only person entitled to this information is the owner of the mailbox.

Express uses a lightweight cryptographic algorithm based on symmetric cryptography. Before changing the contents of a mailbox, the servers must decrypt it and then re-encrypt it. To do this, both servers receive their keys k_A and k_B from the client when creating a mailbox. After each change, all mailboxes are re-encrypted. This means that an attacker cannot find out which mailbox a message has been stored in unless they have both keys.

A counter is used for encryption. So when re-encryption is required, the previous value is subtracted and the new value added, which is a commutative operation. It is not necessary to re-encrypt after every write, but only if a read has occurred between writes. Otherwise, no one, including an attacker, has obtained any information. This increases the efficiency of encryption. In addition, it is not absolutely necessary for both servers to perform the re-encryption. If one of the servers has been corrupted and no longer re-encrypts, the system remains secure.

4.2 Privacy with malicious clients

The techniques presented in the previous chapter only provide security as long as all clients behave honestly. Even a single malicious client can attack and damage both servers. Nothing prevents a client from creating a vector that writes to multiple mailboxes simultaneously. To do this, it creates a DPF whose added vectors

w_A and w_B , calculated from the functions f_A and f_B , are not zero everywhere or in multiple indices. Thus, one message from a malicious client is sufficient to overwrite all mailboxes and thereby delete the original content. Express has introduced an auditing protocol to prevent this. This ensures that each write only updated only one mailbox. Each vector to be inserted must therefore have a zero as an element at every position except one. If the vector does not meet this requirement, it can be discarded, meaning that the write request is rejected by the server and nothing is written to any mailbox.

4.3 Express for Whistleblowing

As explained in the previous chapters, Express offers a secure communication option. Outside of the data transfer itself, there are two more safety issues that should be considered. First, we need plausible deniability, second, we need a way to agree on a Mailbox address.

4.3.1 plausible deniability / cover traffic

If whistleblowers were the only users of Express, it would be obvious that anyone who sent a message over Express had passed on information. This means that many users that are not involved in whistleblowing must also send messages, creating cover traffic. To achieve this, a JavaScript program can be integrated into a website with high traffic, such as a news site. Every time a user visits this website, a dummy request is sent to the Express servers. The servers generate a key that is not known to any client. This means that these messages cannot be read, but they also cannot be distinguished from those of the whistleblowers. News websites like New York Times benefit from high cover traffic because whistleblowers feel safer and are more likely to pass on information. This makes such sites particularly suitable for integrating JavaScript. The paper(ref!) does not specify how cover traffic is required exactly, but it should be enough to ensure that real messages remain indistinguishable from the generated ones. The cost and computation times are very low, which can further encourage the use of Express.

4.3.2 Dialing

To select the mailbox address, the journalist and the whistleblower can meet in person. However, this is often not possible. In this case, a dialing protocol can be used. For example, the Riposte system could be used as a basis and made faster and cheaper with the improvements identified in Express. Journalists post their public keys for Riposte online, then whistleblowers can send them a mailbox address. The Journalist can download their messages and register the mailboxes.

5 Related work

In this section we will present Pung, Riposte and Addra, three cryptographic metadata-hiding communication systems working with the concept of mailboxes. Pung and Riposte were developed before Express and were used as sources in the paper on Express. Addra, on the other hand, was developed later and uses Express as a source.

Pung addresses the same problem as Express, focusing on a solution for whistleblowers, but uses more computationally intensive public key cryptography. Riposte uses similar methods to Express, but does not provide for secret messages. Addra's goal is also to hide metadata(Ahmad et al., 2021). However, this does not concern written messages, but voice calls. Then we will compare their security and performance to Express.

5.1 Pung

Pung (Angel and Setty, 2016) is designed to provide privacy guarantees of both message content and metadata. Its focus lies on protection against powerful global adversaries who may control the entire network infrastructure. This could be mass surveillance or state-sponsored attacks. In Pung, clients deposit and retrieve messages through an untrusted key-value stores. Only one server is required. Pung provides privacy even when the server is malicious.

Pung's main goal is to hide the communication patterns of clients who already know each other's identities. It allows two or more users to exchange messages over a public network without anyone else learning that

they communicated. To prevent network surveillance, Pung operates in rounds. In each round, a client can only send and receive a limited number of messages. If this limit is exceeded, users must wait for the next round. Message retrieval is based on private information retrieval (PIR). As a result, clients have to download large amounts of data simultaneously to ensure anonymity. Pung is therefore particularly suitable for applications where this is the case, such as emails or group chats. Otherwise, the cost of network traffic is relatively high.

While significantly reducing the costs compared to prior approaches, Pung's network costs still remain high. However, newly introduced techniques have significantly reduced the cost of cryptographic calculations.

5.2 Riposte

Riposte protects against network surveillance and malicious clients (Corrigan-Gibbs et al., 2015). Like Express, the main target group is whistleblowers who want to leak information anonymously. Riposte users can post messages anonymously on a bulletin board, which acts as a public broadcast channel. Clients have a fixed time window in which they can write their messages. These are collected on the server and published when the time window has expired. Now everyone can read all the messages written previously. The content is therefore not secret, but the author remains anonymous.

Riposte can be used in two ways. Either with three servers, only one of which may be malicious, or with a variable number, at least one of which must be honest. The variant with three servers, at least two of which are honest, uses a less expensive protocol. It is therefore preferable when millions of clients use Riposte, and it is also the one compared with Express in the paper.

Using DPFs and a private information retrieval (PIR), Riposte minimizes the number of bits each client must upload to each server.

Riposte's key innovation was minimizing the amount of data each client uploads to the servers, making large-scale anonymous writing more practical than earlier systems. However, it relies on public-key cryptography and synchronized communication rounds, which still lead to computational and communication overhead.

5.3 Addra

Addra is dedicated to hiding metadata during voice calls (Ahmad et al., 2021). This data, such as call duration, time, parties involved, can be very interesting for government surveillance programmes and is therefore highly sensitive. Addra is the first system to hide metadata from voice communications over a completely untrusted infrastructure. Previous systems with the same goal were not enough scalable, with a maximum of a few dozen users, or they required trusted partners, which posed a security risk.

Callers receive a mailbox when they initiate a call, regardless of who it is going to. They place their message in it, preventing an enemy from finding out who the call is going to. Using the PIR protocol, the callee can retrieve this message from the mailbox. Addra extends Pung's protocol using public and private keys to establish secure communication channels. Addra uses Pung's PIR protocol with private and public key encryption. However, Addra has expanded it to FastPIR. This reduces calculation times on the server side. In addition, PIR calculations are parallelised, as low latency is essential for calls.

Addra has demonstrated that metadata privacy is also important for more demanding areas than sending messages and can be implemented using similar methods.

5.4 Comparison

All four applications focus on combining scalability with strong privacy guarantees against network surveillance and malicious servers. A significant limitation of Pung and Riposte compared to Express is that both work with batched rounds, meaning each client must have written their message before anyone can read it. This simplifies anonymity guarantees, but restricts usability, since communication can only proceed once all clients in a round have acted. Batch processing introduces high latency and makes real-time interaction impractical.

Express eliminates this limitation through an asynchronous architecture that allows clients to write and read messages independently. This not only improves responsiveness but also reduces the coordination overhead that previously limited scalability. Express therefore offers comparable cryptographic privacy with a better user experience. Riposte focuses on anonymous public posting, functioning as a broadcast channel

rather than private messaging. Pung and Addra on the other hand enable private message exchange between known parties over an untrusted infrastructure. However, both systems achieve this by accepting higher computational and network costs, which limits their scalability to smaller user groups.

5.4.1 Security

In times when technical possibilities are constantly expanding and nation-states have a high interest in metadata, relying on trusted servers is risky. States can use their enormous political and financial resources to attack these servers. That is why Pung and Addra, which continue to operate even without a trusted infrastructure, offer the strongest security. Riposte and Express take a different approach by distributing trust across two or more servers. Compared to Riposte, Express strengthens its security guarantees through an auditing mechanism that detects malformed client requests and prevents data corruption without revealing metadata. This protection against malicious clients makes Express more robust in practical deployments while still maintaining a realistic trust model. Therefore, Express provides a balanced compromise between strong cryptographic privacy and operational practicality, making it more suitable for large-scale or real-world use.

5.4.2 Performance

For a fixed security parameter λ , Express' auditing protocol requires $O(\lambda)$ communication between parties and $O(1)$ client side communication. Riposte on the other hand required $\Omega(\lambda\sqrt{n})$ communication and $\Omega(\sqrt{n})$ client computation. In addition, Riposte requires three servers instead of two like Express. This results in the new auditing scheme reducing the costs for the client by $8\times$ for an installation with one million registered mailboxes.

Significant events prior to the research of Express have led to Express testing larger messages than previous systems. In 2019, a whistleblower submitted a report about President Trump and his interactions with the leader of Ukraine (Anonymous, 2019). The file had a size of 25.3 KB and showed how important it is to offer whistleblowers the option of sending long messages.

Overall Express reduces the costs of maintaining a service for whistleblowers by $6\times$ compared to previous work with the same objective.

6 Conclusion

We have shown that Express made progress in performance and reproducibility of privacy enhancing technologies. Its auditing and virtual addressing achieve efficient, scalable metadata privacy suitable for whistleblowing, journalism, or any communication system where metadata leakage is a risk. We also compared Express with three other systems that protect metadata from attackers. By comparing the calculation costs and computation times of the different applications, we found that Express became more cost-effective and time-efficient without sacrificing privacy guarantees.

7 Use of generative digital tools

The use of AI tools must always be declared completely and transparently. For examples of how to do this, please see the DMI guidance: https://dmi.unibas.ch/fileadmin/user_upload/dmi/Studium/Computer_Science/Diverses/Wegleitungen/Dokumente_2023/Guidance_on_AI_in_teaching_in_Computer_Science.pdf.

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